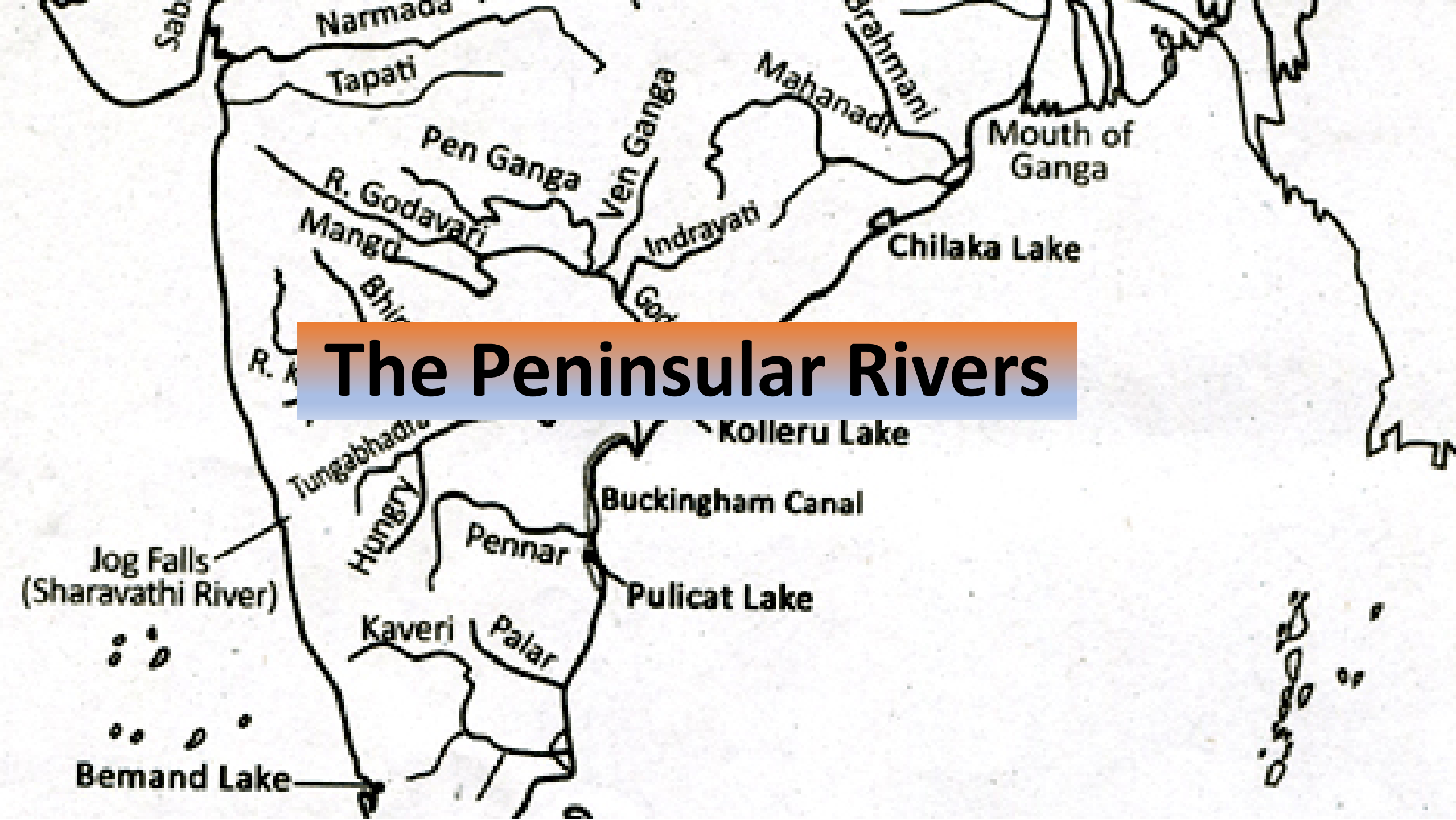


The Peninsular Rivers

Evolution of Peninsular Drainage System

- Three major geological events in the distant past have shaped the present drainage systems of Peninsular India
- Subsidence of the western flank of the Peninsula leading to its submergence below the sea during the early tertiary period. Generally, it has disturbed the symmetrical plan of the river on either side of the original watershed.

- A watershed is an area of land that drains or “sheds” water into a specific waterbody. Every body of water has a watershed. Watersheds drain rainfall and snowmelt into streams and rivers. These smaller bodies of water flow into larger ones, including lakes, bays, and oceans. Gravity helps to guide the path that water takes across the landscape.

- Upheaval of the Himalayas when the northern flank of the peninsular block was subjected to subsidence and the consequent trough faulting. The Narmada and The Tapti flow in trough faults and fill the original cracks with their detritus materials. Hence, there is a lack of alluvial and deltaic deposits in these rivers.

- Slight tilting of the peninsular block from northwest to the south-eastern direction gave orientation to the entire drainage system towards the Bay of Bengal during the same period.

- The Peninsular drainage system is older than the Himalayan one. This is evident from the broad, largely-graded shallow valleys, and the maturity of the rivers. The Western Ghats running close to the western coast act as the water divide between the major Peninsular Rivers, discharging their water in the Bay of Bengal and as small rivulets joining the Arabian Sea

- Most of the major Peninsular Rivers except Narmada and Tapti flow from west to east. The Chambal, the Sind, the Betwa, the Ken, the Son, originating in the northern part of the Peninsula belong to the Ganga river system. The other major river systems of the peninsular drainage are – the Mahanadi the Godavari, the Krishna and the Kaveri. Peninsular rivers are characterised by fixed course, absence of meanders and non-perennial flow of water. The Narmada and the Tapti which flow through the rift valley are, however, exceptions.


Indus
Jhelum
Ravi
Chenab
Sutlej
Ganga
Yamuna
Brahmaputra
Narmada
Tapti
Mahanadi
Godavari
Krishna
Kaveri
Arabian Sea
Bay of Bengal

- **The Godavari**: It is the largest peninsular river system due to this it is also called the Dakshin Ganga.
- It rises in the Nasik district of Maharashtra and discharges its water into the Bay of Bengal. Its tributaries run through the states of Maharashtra, Madhya Pradesh, Chhattisgarh, Orissa and Andhra Pradesh.

- The Penganga, the Indravati, the Pranhita, and the Manjra are its principal tributaries. The Godavari is subjected to heavy floods in its lower reaches to the south of Polavaram, where it forms a picturesque gorge. It is navigable only in the deltaic stretch. The river after Rajamundri splits into several branches forming a large delta.

- The Godavari is India's second longest river after the Ganga. Its source is in Triambakeshwar, Maharashtra. It flows east for 1,465 kilometers (910 mi), draining the states of Maharashtra (48.6%), Telangana (18.8%), Andhra Pradesh (4.5%), Chhattisgarh (10.9%) and Odisha (5.7%).

- The river ultimately empties into the Bay of Bengal through an extensive network of tributaries. Measuring up to 312,812 km2 (120,777 sq mi), it forms one of the largest river basins in the Indian subcontinent, with only the Ganga and Indus rivers having a larger drainage basin. In terms of length, catchment area and discharge, the Godavari is the largest in peninsular India, and had been dubbed as the Dakshin Ganga (Ganges of the South).

- The Krishna River is the fourth-biggest river in terms of water inflows and river basin area in India, after the Ganga, Godavari and Brahmaputra. The river is almost 1,288 kilometres long.
- The river is also called Krishnaveni. It is one of the major sources of irrigation for Maharashtra, Karnataka, Telangana and Andhra Pradesh.

- The Krishna river originates in the Western Ghats near Mahabaleshwar at an elevation of about 1,300 metres (4,300 ft), in the state of Maharashtra in central India.). It is one of the most suitable arable basin as 75.6 % area of the Krishna basin is under cultivation because of availability of water.

- The river's source is at Mahabaleshwar near Jor village in the extreme north of Wai Taluka, Satara District, Maharashtra in the west and empties into the Bay of Bengal at Hamasaladeevi (near Koduru) in Andhra Pradesh, on the east coast. It flows through the state of Karnataka before entering Telangana State. The delta of this river is one of the most fertile regions in India and was the home to ancient Satavahana and Ikshvaku dynasty kings. Vijayawada is the largest city on the Krishna.

- The largest tributary of the Krishna River is the Tungabhadra River with a drainage basin measuring 71,417 km2 (27,574 sq mi), running for about 531 km (330 mi), but the longest tributary is the Bhima River, which makes a total run of 861 km (535 mi) and has an equally large drainage area of 70,614 km2 (27,264 sq mi).

- Kaveri is an Indian river flowing through the states of Karnataka and Tamil Nadu. The Kaveri river rises at Talakaveri in the Brahmagiri range in the Western Ghats, Kodagu district of the state of Karnataka, at an elevation of 1,341 m above mean sea level and flows for about 800 km before its outfall into the Bay of Bengal.

- It is the third largest river – after Godavari and Krishna – in South India and the largest in the State of Tamil Nadu, which, on its course, bisects the state into North and South. The Kaveri is sacred river to the people of South India and is worshipped as the Goddess Kaveriamma. The Kaveri is also one of the seven holy rivers of India

- The catchment area of Kaveri basin is estimated to be 81,155 square kilometres (31,334 sq mi) with many tributaries including Harangi, Hemavati, Kabini, Bhavani, Lakshmana Tirtha, Noyyal and Arkavati. The river basin covers three states and a Union Territory as follows: Tamil Nadu, 43,868 square kilometres ; Karnataka, 34,273 square kilometres ; Kerala, 2,866 square kilometres , and Puducherry, 148 square kilometres